ITAI-1371 - Intro to Machine Learning

Individual Contribution Journal

Camila Ferreira da Silva

My contribution to Lab 8 included work on Parts 2 and 3. In Part 2, I fixed a formatting issue in the synthetic-data setup cell: the comment # Generated some synthetic data and np.random.seed(0) had been joined incorrectly, causing a NameError. I separated them onto their own lines so the data setup could run correctly. In Part 3, I corrected the indentation inside the for loop so the code could create, train, make predictions with, and plot the polynomial models. I also added the necessary imports and defined X and y in the cell so it would not depend on variables from earlier cells after a runtime reset. These changes addressed the errors that had prevented the code from running. I also answered Knowledge Check question 1, explaining that degree 1 underfits, degree 4 provides the best fit, and degree 15 overfits the data. I also integrated the group members' sections and ran the notebook from top to bottom to check that the cells executed in order and produced the expected outputs.

Danh Nguyen

I was responsible for the degree 15 polynomial model. I built and trained the model, plotted its predictions against the data and the true sine-wave pattern, and calculated its training error and 5-fold cross-validation error to show the overfitting gap (about 0.21 vs. 1.03). I added feature scaling to keep the model numerically stable, and I wrote the explanation of why degree 15 overfits (high variance).

Great Inioluwa Olayiwola

My main contribution was writing the missing Python code for the construction, training, and visualization of the polynomial regression models and their learning curves. To do this, I first built pipelines using Scikit-Learn to iterate over degrees 1, 4, and 15, and to plot their predictions against generated data. Then, I wrote the plot_learning_curve function for both the degree 1 (underfit) and degree 15 (overfit) cases, with cross-validation set to 5. I then supplemented this code by writing analytical answers for the questions, explaining how complexity influences fit, the interpretation of the trends in the learning curves (i.e., bias/variance trade-off), and how this influences the choice of model.

Karima Sadruddin

For this module, I worked on building and plotting a Degree 4 Polynomial Regression model using Python. I used the PolynomialFeatures and LinearRegression functions to create and train the model. I also generated predictions and plotted the model against the original data points to observe how well it followed the overall pattern.

I compared the Degree 4 model with simpler and more complex models to understand underfitting and overfitting. I also reviewed the model's performance and its R^2 score, which helped me understand how much variation the model could explain.

While completing this task, I faced some challenges understanding how polynomial degrees affect the shape and accuracy of a model. It took some time to understand why a Degree 4 model could provide a better balance than a model that was too simple or too complex. However, by studying the graphs and reviewing the results, I was able to understand the concept better and complete the task.

Overall, my contribution was building the Degree 4 model, generating its graph, analyzing the results, and explaining why it represents a reasonable balance between underfitting and overfitting.

Quan Thach

For this lab, I completed Task 2 by building the learning curve for the underfit model. I took the plot_learning_curve function, which had been flattened onto a single line and wouldn't run, and rewrote it with proper formatting so it worked correctly. I then created the degree 1 polynomial regression pipeline and plotted its training and cross-validation scores against the number of training examples. I also wrote the answer to Knowledge Check question 2, explaining that the low, closely clustered scores that plateau early point to high bias, meaning the model is too simple for the sine-wave pattern and more data alone would not fix it.